☆ Example of sliding window.

set : x = [ [ 1, 2, 3, 4]
[ 5, 6, 7, 8]
[ 9, 10, 11, 12]
[13, 14, 15, 16] ]

kernel size KH = KW = 2. it selects a small 2x2 window each time.

How should I select the first (0,0) small window?

The top-left corner of the window is at (0,0)

So select x[0:0+2, 0:0+2] → is x[0:2, 0:2]

x[row_start : row_end, col_start : col_end]

result is → [ [1, 2]
[5, 6] ]

move the window one step to the right →

The top-left corner of second window becomes (0,1).

x[0: 2+0, 1: 2+1] → x[0 : 2, 1:3] → [ 2, 3
6, 7 ]

Continue to go right. $X[0:2, 2:4] \rightarrow \begin{bmatrix} 3, 4 \\ 7, 8 \end{bmatrix}$.

So there are 3 windows in the first line.

Now the window slides down one line. Now the top-left corner is at (1, 0).

$X[1:3, 0:2] \rightarrow \begin{bmatrix} 5, 6 \\ 9, 10 \end{bmatrix}$ continue $X[1:3, 1:3]$

$\rightarrow \begin{bmatrix} 6, 7 \\ 10, 11 \end{bmatrix}$ continue $X[1:3, 2:4]$ $\begin{bmatrix} 7, 8 \\ 11, 12 \end{bmatrix}$

So how many windows in total?

Input size 4x4
kernel 2x2
padding = 0
stride = 1

$$H_{out} = \frac{H + 2P - KH}{S_{=1}} + 1 = \frac{4+0-2}{1} + 1 = 3$$

$$W_{out} = \frac{W + 2P - KW}{S} + 1 = \frac{4+0-2}{1} + 1 = 3$$

So there are 9 small windows in total, these correspond to the 9 locations in the output feature map.

⚠ Don't include the right boundary

e.g.: $[2:\cancel{4}] \rightarrow [2:3]$ ✓

einsum.

result $= \Sigma_h \Sigma_w$ window$[h, w] \times$ kernel$[h, w]$

e.g.: window = np.array([
[1, 2],
[5, 6]
])

kernel = np.array([
[1, 0],
[0, 1]
]).

result = np.sum(window * kernel)

$$= [[1*1, 2*0, 5*0, 6*1]] = [[1,0], [0,6]]$$

$1+0+0+6=7$ so finally output result = 7